

Abstract

Distribution-free certification of learned progressive policies for retrieval-augmented generation under knowledge drift

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Dynamic retrieval-augmented generation (RAG) learns when a language model should retrieve, rerank, or escalate to a stronger generator. These policies can improve the accuracy–cost trade-off, but their scores do not bound population risk and may become unreliable after knowledge drift. We introduce **CONTRACTRAG**, a learning-and-certification method for progressive RAG inference. Candidate policies use learned sufficiency scores over retrieval and generation signals to stop at the cheapest adequate rung. Risk vouchers summarize the observed effect of plan rewrites and order a large policy family. Soundness comes from a separate distribution-free fixed-sequence test: with probability at least $1 - \delta$, every loss in the selected evidence contract meets its target, regardless of candidate count or score quality. We also prove a cost-consistency result and give a restart-mixture e-process whose drift-detection delay is independent of the change time. A geometric schedule preserves validity through repeated recertification. Experiments on HybridQA, CRAG, ASQA, and QAMPARI cover four model families, two serving backends, and ablations of scoring, policy families, calibration size, candidate density, audit rate, and drift. Across 1000 calibration draws, selected policies violate their contracts in at most 0.8% of draws under a 10% error budget, whereas empirical and Bayesian selection reach 64%. The policies reduce model cost by up to $20\times$ ($10\text{--}18\times$ with audits). The method is orthogonal to the underlying retriever or reranker and provides a general certification layer for learned adaptive RAG.

Keywords: Retrieval-augmented generation; Adaptive computation; Distribution-free risk control; Uncertainty estimation; Drift detection; Large language models